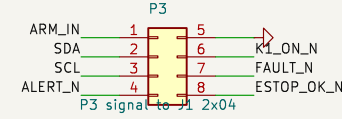
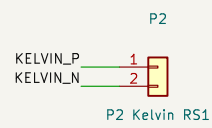
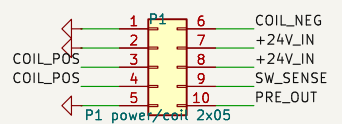
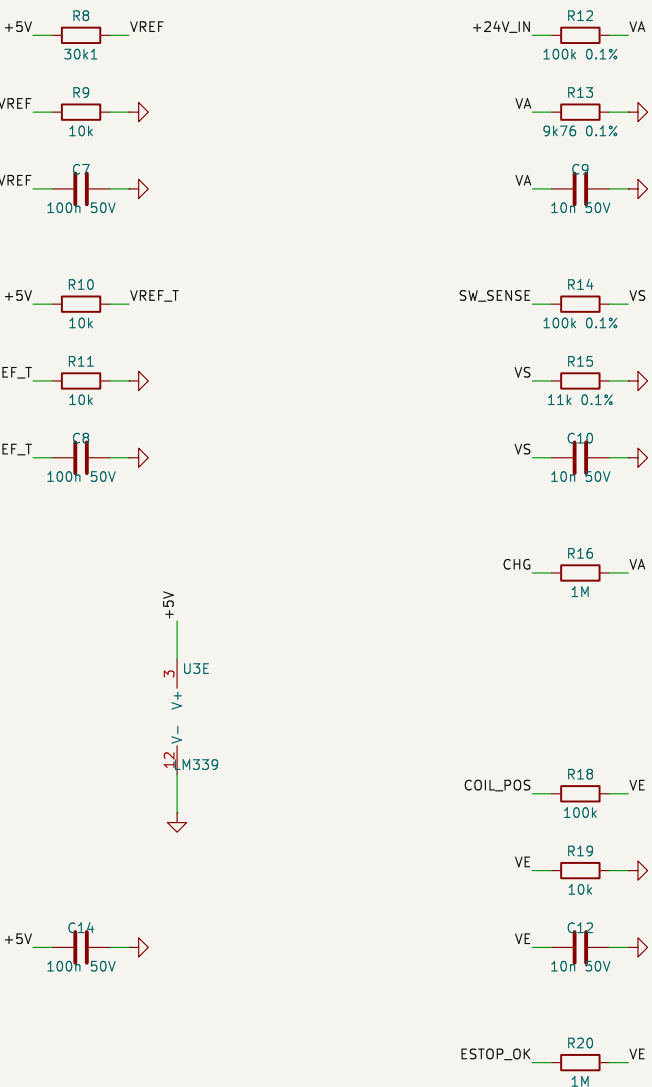


CONNECTORS — Molex Micro-Fit 3.0, vertical



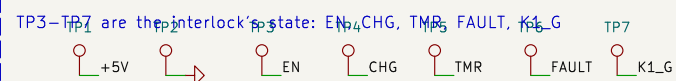
P1 is on the board's TOP edge (towards K1 and R1), P2 on the LEFT (towards RS1), P3 on the RIGHT (towards J1) — panel_interface.json says so and check_fit.py checks it. The E-stop loop is on P1, not P3: it is a 2 A coil circuit. Pins 1–5 are the row nearest the board edge: GND and the E-stop return / coil+ pair, which need no route out. Every net that has to

comparators for LM339, open-collector, 5 V



A PRE_OK: switched bus / always-on bus > 90 % A RATIO, so it holds from 21 V to 29.4 V. R12–R15 are 0.1 %: R16 is the hysteresis, and because it injects 5 V into a divider of a varying pack the threshold moves 88.3 .. 91.7 % over every corner — below the 94.5 % the bus can reach through R1 with the ESCs idling. B FAULT timer: CHG charges C11 through R17; 1.7 s minimum against a 0.73 s worst precharge. D4 latches. C ESTOP_OK: the far side of the E-stop loop is above 13.7 V. D ARM: host logic above 1.37 V (LVTTL). Every output node is pulled up to EN, not to 5 V, wherever it feeds the interlock: disarm and they all fall.

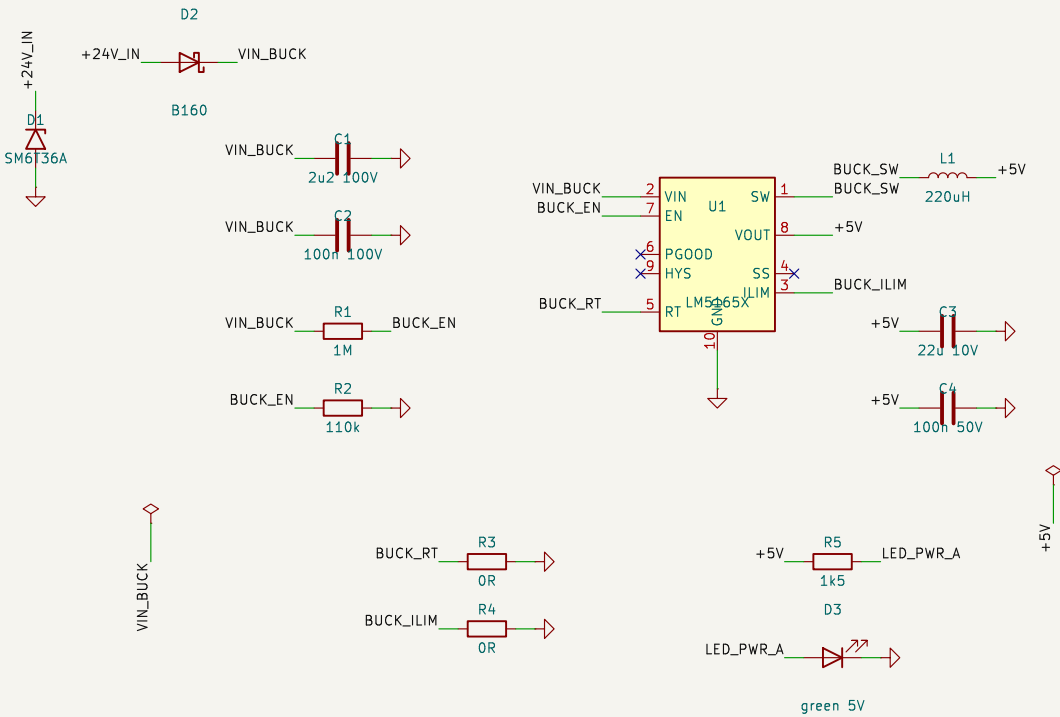
TEST POINTS + MOUNTING



H1–H4: M3, positions from panel_interface.json

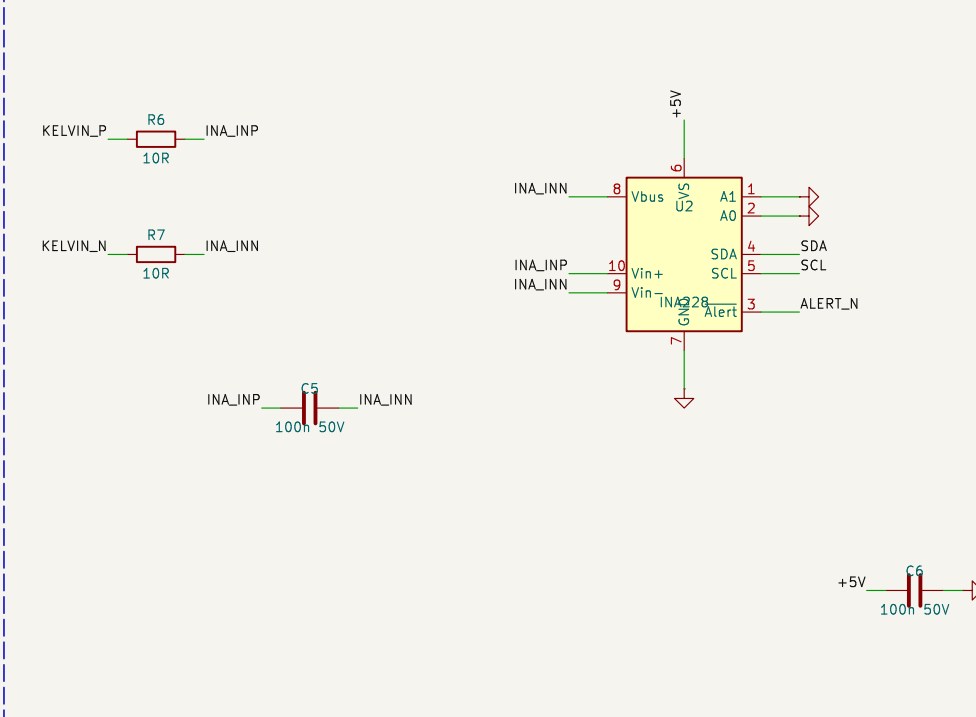


24 V INPUT PROTECTION + 5 V SUPPLY



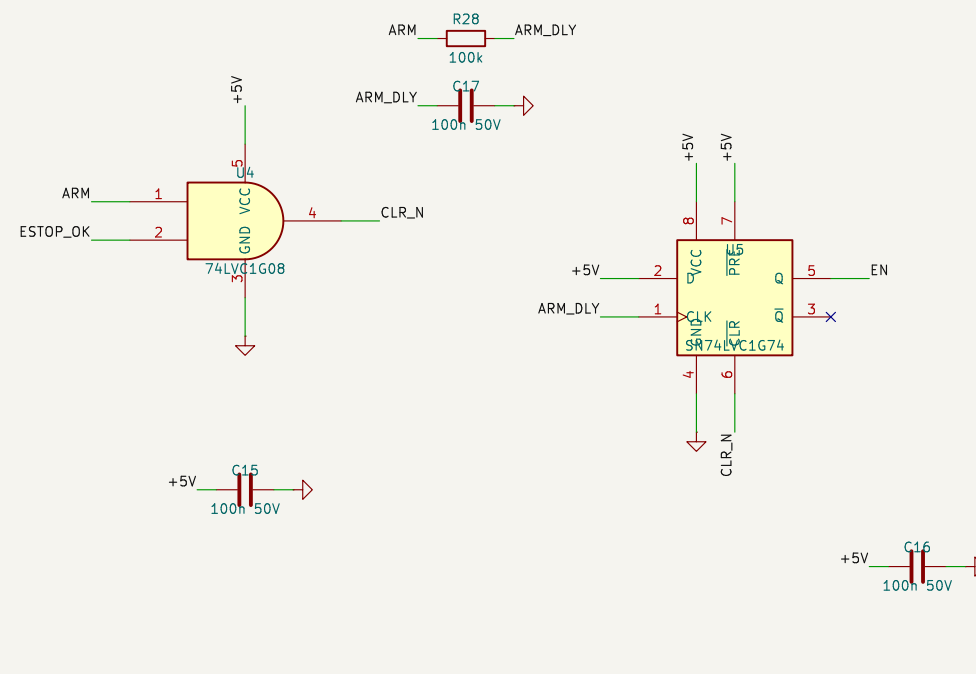
D1 stands off 30.8 V (a full 7S pack is 29.4 V) and clamps at 49.9 V, under the 60 V of every FET on the 24 V side. It is unidirectional on purpose: a reversed supply forward-biases it and blows the panel's 2 A fuse at DIN2–203. D2 keeps a reversed pack off the buck in the meantime. R1/R2 hold the buck off below 12 V. R3 and R4 are OR straps — see spec.VERIFY: RT and ILIM to GND select PFM and the low current limit.

PACK CURRENT — INA228 on panel shunt RS1 (0.75 mOhm)



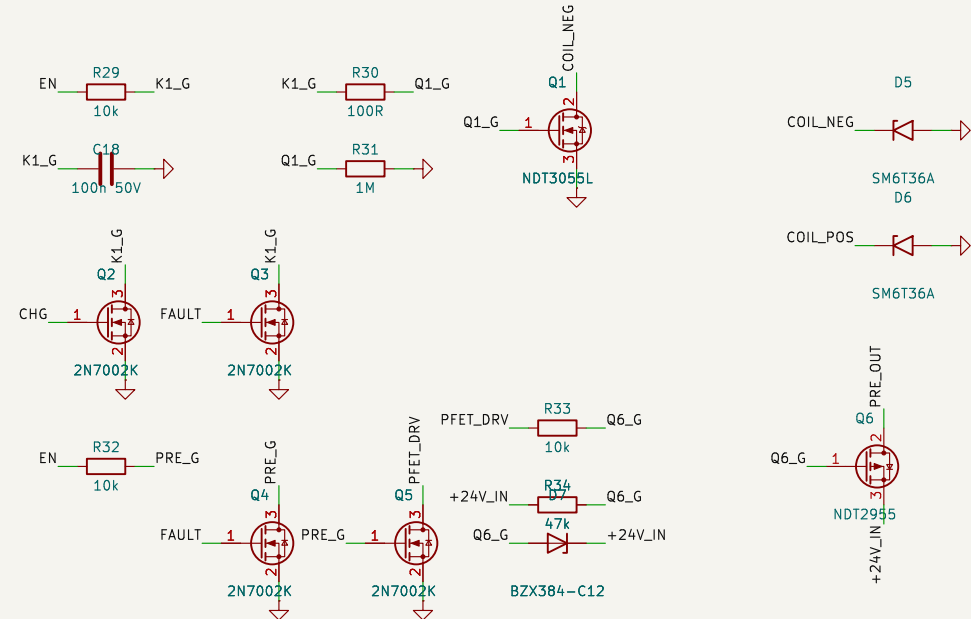
ADCRANGE = 0: +/-163.84 mV = +/-218 A full scale, 0.42 mA per count. SHUNT_CAL = 2803 for CURRENT_LSB 286.1 uA (design/calcs.py). R6/R7/C5 are TI's input filter — 10 ohm is the most the datasheet allows before bias current becomes gain error. VBUS reads the load side of the shunt, which is the always-on bus. Address 0x40 (A0 = A1 = GND). No I2C pull-ups here: the host supplies them at ITS logic level.

INTERLOCK LATCH — EN = armed by a fresh ARM edge while the E-stop loop is closed



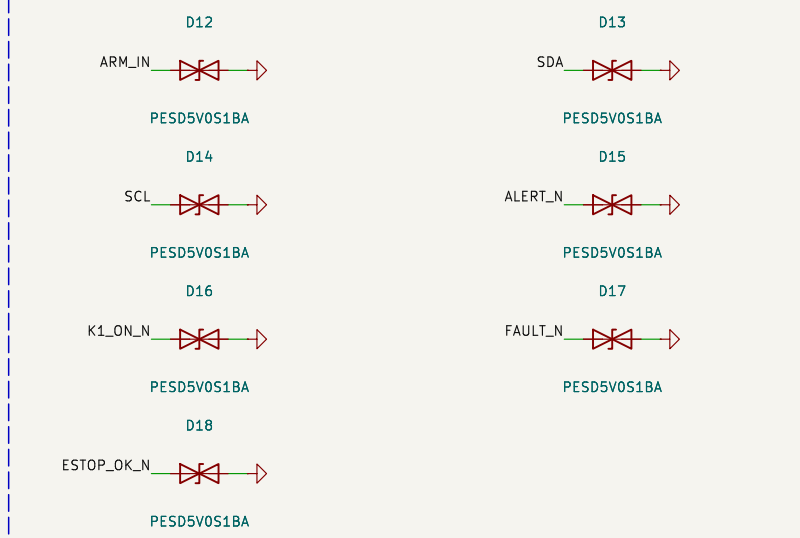
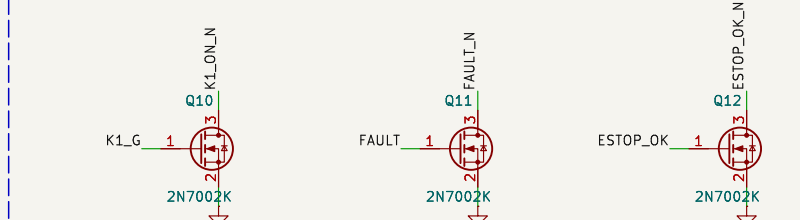
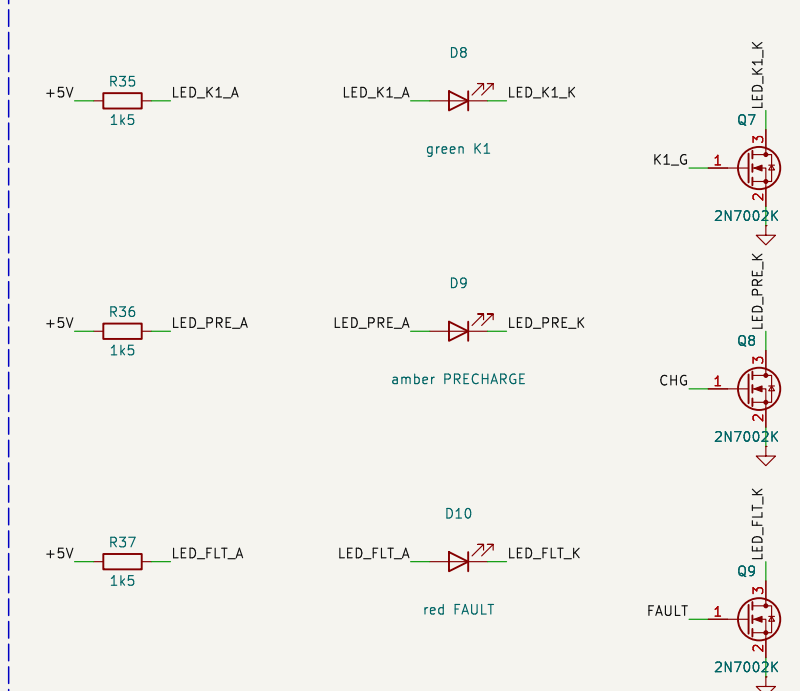
U4 makes CLR# = ARM AND ESTOP_OK, which clears U5 the instant either drops. U5 only SETS on a rising clock edge, and the clock is ARM delayed 10 ms by R28/C17 so CLR# is always released first. The consequence is the one the rules ask for: releasing the E-stop does not re-energise K1 on its own — the host has to drop ARM and raise it again.

K1 COIL DRIVE + PRECHARGE SWITCH



K1.G = EN, pulled low by Q2 while precharging and by Q3 on FAULT; C18 makes it rise 1 ms after CHG falls, so a glitch cannot reach the coil. The coil current path is +24V -> P1.8 -> E-stop -> P1.3 -> P1.4 -> coil -> P1.6 -> Q1 -> GND; Q1 can only interrupt it. D5/D6 clamp both coil wires, which run off the board. Q6 switches pack -> R1 -> switched bus; D7 clamps its gate to 12 V from 21 V to 29.4 V.

STATUS — LEDs at the hatch, open-drain to the host



Every output to J1 is an open-drain 2N7002 (active low); the host pulls up to its own logic level, so this 5 V board never drives a 3.3 V pin. D12–D18 are on every line that leaves the panel.

Generated by hw/gen/gen_sch.py from a pin -> net table. Edit the generator, not this file. The E-stop loop is in series with the K1 coil. This board can open it; nothing on it can bypass it. K1 can close only if: E-stop loop closed AND a fresh ARM edge AND switched bus >= 90 % of pack.

Sheet: /
File: rover-panel-supervisor.kicad_sch
Title: rover-panel-supervisor — precharge, K1 interlock, pack current
Size: A2 Date: Rev: A
KiCad E.D.A. 9.0.5 Id: 1/1